

A positive-characteristic counterexample to finite algebraic Banach field extensions

Automated Math Discovery Run `fa_banach_001`
Model: GPT5.6

22 July 2026

Status. Counterexample likely valid, subject to human review. This gives a full negative answer to the exact question in Remark 7.3 of arXiv:2504.10209v1; it does not settle the paper’s broader positive-characteristic automatic-continuity problem.

1 Source question

Feliks Rączka’s paper *Continuity of differential operators for nonarchimedean Banach algebras* proves the following as Lemma 7.2 in characteristic zero: if $K \subset L$ is algebraic, K is a nontrivially valued nonarchimedean field, and L is a Banach K -algebra which is a field, then $[L : K] < \infty$ and L is nonarchimedean [1]. Remark 7.3 asks whether this remains true when $\text{char } K = p > 0$. It specifically observes that purely inseparable extensions defeat the root-separation argument used in characteristic zero.

Remark 7.3. We do not know if the above lemma holds when $\text{char } K = p > 0$. Note that if the extension $K \subset L$ is purely inseparable then the value (7.4) is not well-defined and hence the above proof does not carry to the positive characteristic. In [7] Lemma 1, Section 3.4.3] it is shown in any characteristic that if the extension $K \subset \overline{K}$ is infinite then $\overline{K}$ is not nonarchimedean with respect to the unique multiplicative nonarchimedean norm extending the one on K . The proof is reduced to separable extensions by observing that if $K \subset \overline{K}$ is infinite then so is $K \subset K_{\text{sep}}$, where K_{sep} is the separable closure of K .

Figure 1: Remark 7.3 on page 11 of the source paper.

2 Definitions and notation

For a field F of characteristic p , write

$$F^p = \{x^p : x \in F\}.$$

This is a subfield. An extension E/F is *purely inseparable of exponent one* if $x^p \in F$ for every $x \in E$. In particular, such an extension is algebraic, although its total degree need not be finite.

For a coefficient field k and an indeterminate t , the Laurent-series field $k((t))$ is equipped with the t -adic absolute value

$$\left| \sum_{n \geq n_0} a_n t^n \right| = q^{\min\{n : a_n \neq 0\}}, \quad 0 < q < 1.$$

It is a complete nontrivially valued nonarchimedean field.

3 Proof intuition

In positive characteristic, Frobenius provides exactly the phenomenon that the characteristic-zero proof cannot see. Start with a complete field L whose coefficient field has infinitely many independent imperfection directions, and use the Frobenius image $K = L^p$ as the base field. The base remains complete, while every element of L becomes algebraic of degree at most p over K . There is nevertheless no uniform bound on the dimension of the whole extension: finitely many imperfection variables already produce p^N independent monomials for arbitrary N .

4 Counterexample

Theorem 1. *For every prime p , there are fields $K \subset L$ of characteristic p such that:*

1. *K is a complete nontrivially valued nonarchimedean field;*
2. *L is a field and a Banach K -algebra for a multiplicative nonarchimedean norm;*
3. *L/K is purely inseparable of exponent one, hence algebraic;*
4. *$[L : K] = \infty$.*

Consequently, the characteristic- p analogue of Lemma 7.2 in arXiv:2504.10209 is false. It is false even when the Banach algebra norm on L is already multiplicative and nonarchimedean.

Proof. Let $u_1, u_2, \dots$ be algebraically independent over $\mathbb{F}_p$, and put

$$k = \mathbb{F}_p(u_1, u_2, \dots), \quad L = k((t)).$$

Give L its t -adic absolute value, normalized by $|t| = q$ for some $0 < q < 1$, and set

$$K = L^p.$$

We first identify K . Frobenius is continuous for the t -adic topology, and in characteristic p it gives

$$\left(\sum_{n \geq n_0} a_n t^n \right)^p = \sum_{n \geq n_0} a_n^p t^{pn}.$$

Conversely, every coefficient in k^p has a unique p th root in k . Therefore

$$K = L^p = k^p((t^p)) = \mathbb{F}_p(u_1^p, u_2^p, \dots)((t^p)). \tag{1}$$

Upon writing $s = t^p$, the right-hand side is the Laurent-series field $k^p((s))$, with $|s| = q^p$. Thus K is complete, nontrivially valued, and nonarchimedean. The same facts hold for L , and multiplicativity gives $|ax| = |a||x|$ for $a \in K$ and $x \in L$. Hence L is a Banach K -algebra.

For every $x \in L$, we have $x^p \in K$, so x is a zero of

$$X^p - x^p \in K[X].$$

Thus L/K is purely inseparable of exponent one and, in particular, algebraic.

It remains to prove that its degree is infinite. Fix $N \geq 1$ and consider the p^N elements

$$u_1^{e_1} \cdots u_N^{e_N}, \quad 0 \leq e_i < p.$$

We claim that they are linearly independent over K . Suppose that a finite relation

$$\sum_{0 \leq e_1, \dots, e_N < p} c_{e_1, \dots, e_N}(t^p) u_1^{e_1} \cdots u_N^{e_N} = 0 \quad (2)$$

holds, with coefficients in $K = k^p((t^p))$. Expand each coefficient as a Laurent series in t^p . Comparing the coefficient of every power t^{pm} in (2) gives a relation over k^p among the same monomials.

Those monomials are linearly independent over k^p . Indeed, any proposed finite relation uses only finitely many of the variables. In the resulting finite rational-function field, the monomials whose exponent in each variable lies in $\{0, \dots, p-1\}$ form a basis over the subfield of p th powers: this follows by adjoining the variables one at a time, each satisfying $X^p - u_i^p$ and contributing degree p . Hence every Laurent-series coefficient in (2) is zero, proving the claim.

We have shown that $[L : K] \geq p^N$ for every N . Consequently $[L : K] = \infty$, completing the counterexample. $\square$

5 What the counterexample does and does not show

The example violates the finiteness conclusion of the proposed characteristic- p lemma. It does not violate the other conclusion: L is itself nonarchimedean. Thus the obstruction is not a pathological submultiplicative Banach norm; it is the unbounded p -rank permitted by pure inseparability.

This packet does not give a full positive-characteristic characterization of automatic continuity for differential operators, nor a replacement for the source paper's Theorems 1.2–1.3. It settles only the precise question in Remark 7.3.

6 Verification and novelty status

The proof was audited along four possible failure points: the exact Frobenius image in (1), completeness of K , algebraicity of L/K , and infinite K -linear dimension. Each is proved explicitly above. No computation is used.

On 22 July 2026, a bounded novelty check searched the run's lightweight indexes and arXiv/web results using arXiv:2504.10209, the exact paper title, the opening phrase of Remark 7.3, the source label `NormsOnFiniteExtensions`, the author, and the terms *purely inseparable*, *p-degree*, and *algebraic Banach field extension*. The search found the source paper and summaries of it, but no separate work claiming to answer Remark 7.3. This bounded search cannot certify novelty; human literature review is still required.

7 Human-review recommendation

Review (1) and the coefficientwise independence step in (2). Once those two points are accepted, all source hypotheses and the failure of finite degree are immediate. Recommended verdict: **likely valid**, with a separate expert novelty check before external dissemination.

References

- [1] F. Rączka, *Continuity of differential operators for nonarchimedean Banach algebras*, arXiv:2504.10209v1 (2025), especially Lemma 7.2 and Remark 7.3. <https://arxiv.org/abs/2504.10209>